COMPSYS302

Design: Software Practice

Introduction to AI #5 – Dataset

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1. Neural Network

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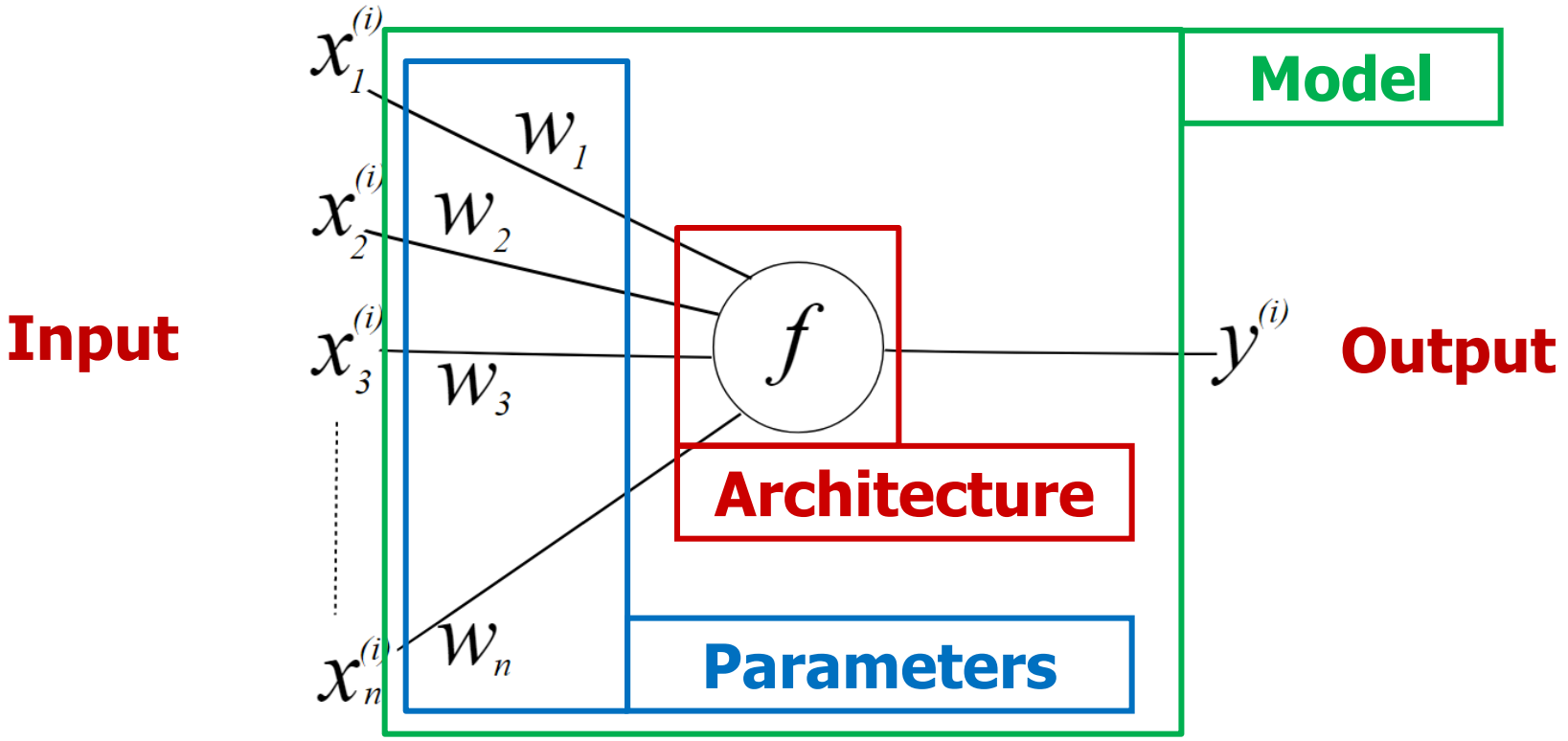
- What is Model?



Model = Architecture + Parameters

2. Concept of CNN

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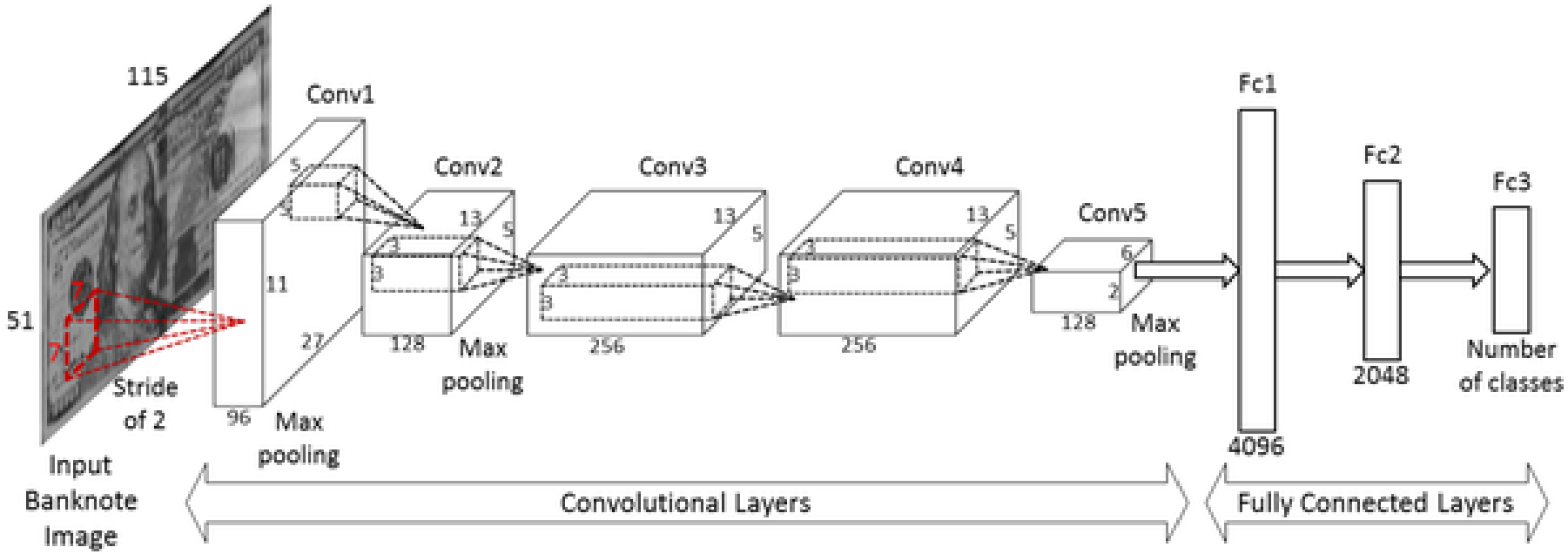
☐ **CNN**



- How deep is convolutional layers?
- Size of kernel matrix for convolutions?
- Which polling methods?
- Which activation functions?

...

Makes different models

2. Concept of CNN

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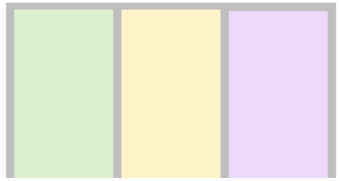
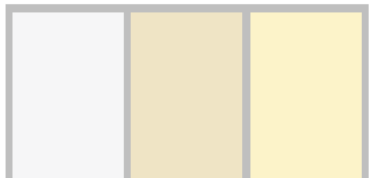
☐ Recap - CNN Convolution

input

CONV ReLU POOL

Convolution





0.02
0.93
0.04

3	1	1	2	8	4
1	0	7	3	2	6
2	3	5	1	1	3
1	4	1	2	6	5
3	2	1	3	7	2
9	2	6	2	5	1

Original image 6x6

"Convolution"

✖

1	0	-1
1	0	-1
1	0	-1

Filter 3x3

Kernel matrix

=

-7	...		
...	...		

Output 4x4

element-wise product
e filter matrix
and the original image

2. Concept of CNN

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☐ Recap - CNN Filtering

input

CONV ReLU POOL

Filtering

CONV ReLU

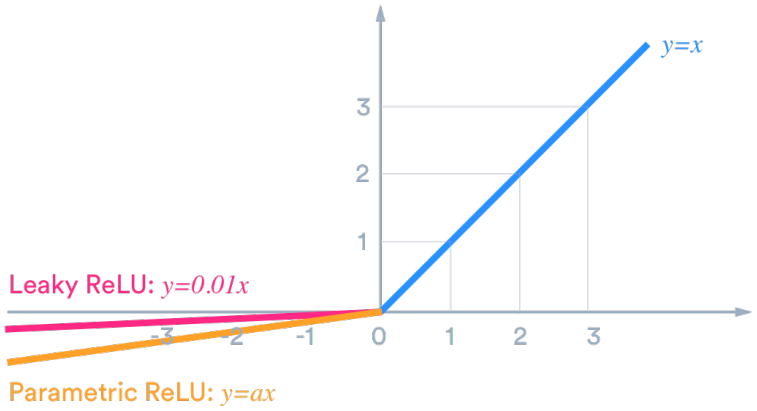
1	0	3	0
4	0	0	0
30	2	0	1
1	0	0	7

output

0.02
0.93
0.04
⋮
0.07
0.11
0.09

"ReLU forward"

1	-2	3	-4
4	-1	-1	-2
30	2	0	1
1	-2	-9	7



[Sringenberg & Dosovitskiy et al. (2015): Visualizing and Understanding Convolutional Networks]

2. Concept of CNN

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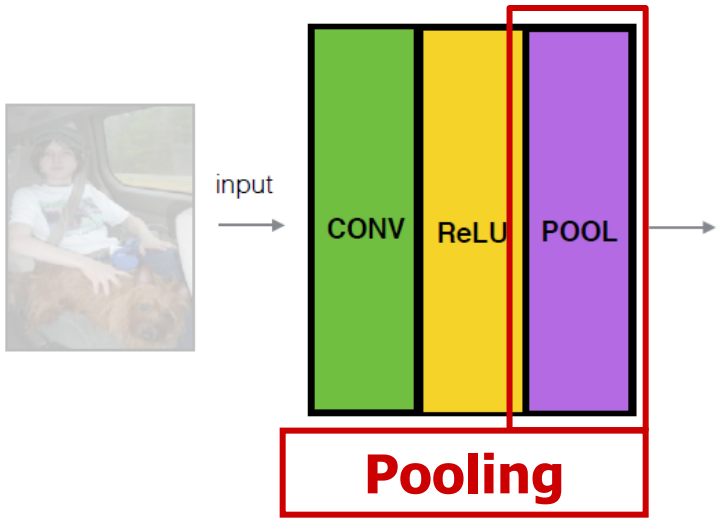
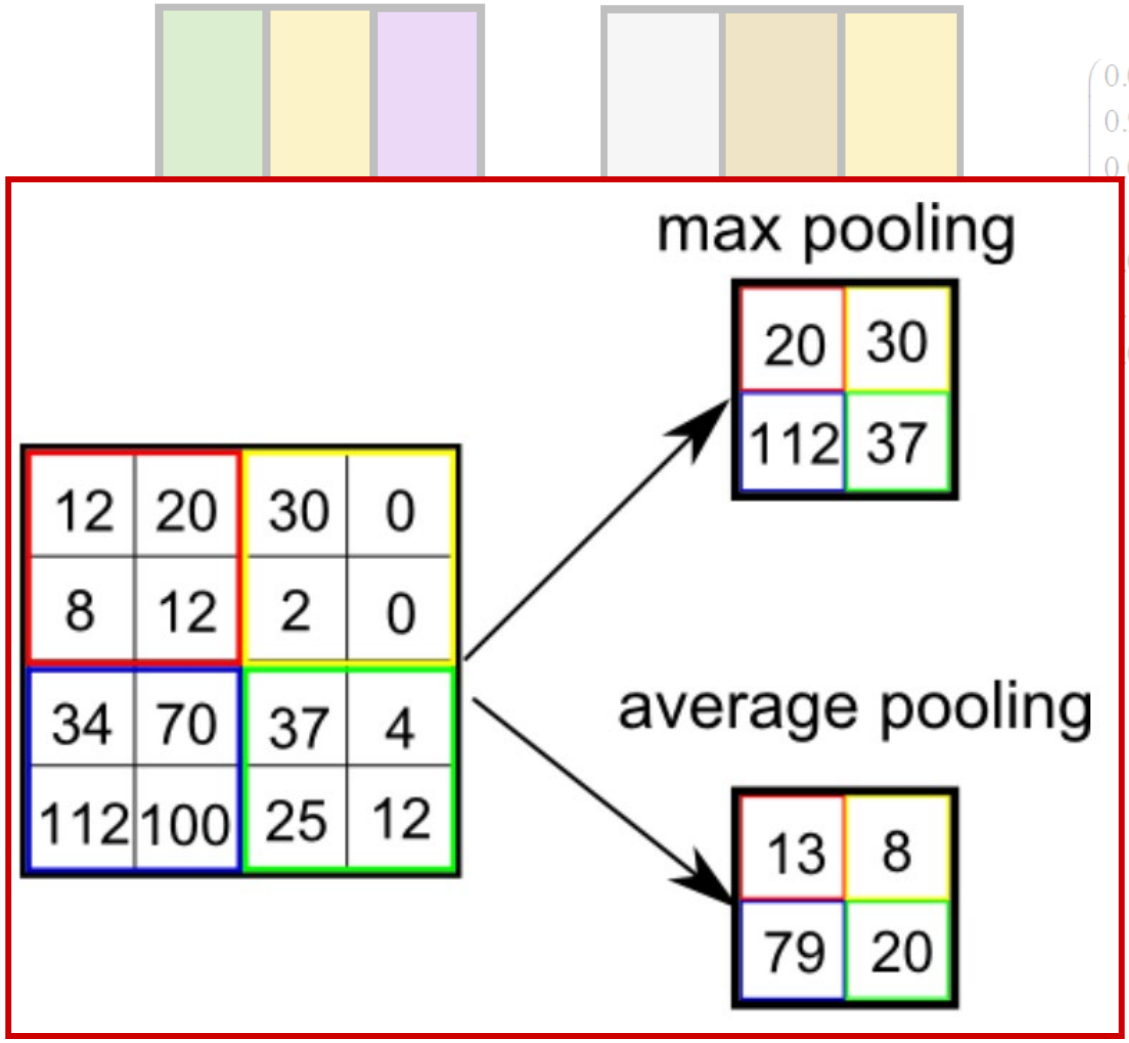
☐ Recap - CNN Polling





3. Different models for CNN

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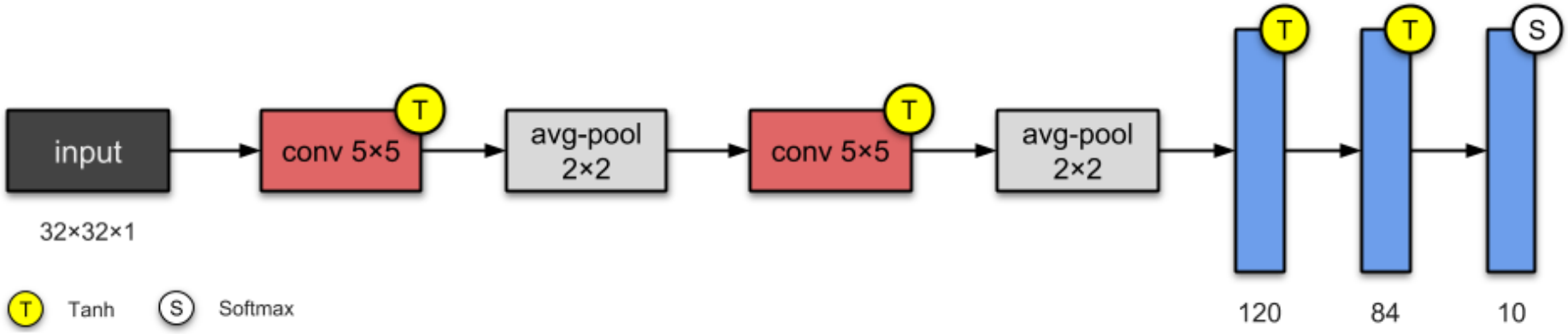
- **LeNet-5 (1998)**
  - one of the **simplest** architectures.
  - has 3 convolutional and 2 sub-sampling layers. ➔ named as "5"
  - has about 60K parameters.
  - The average-pooling layer as we know it now was called a "sub-sampling layer".

  ➔ This architecture has become the **standard 'template'**: **stacking convolutions and pooling layers, and ending the network with one or more fully-connected layers**.



[Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner, "Gradient-Based Learning Applied to Document Recognition," Proceedings of the IEEE, 1998]

3. Different models for CNN

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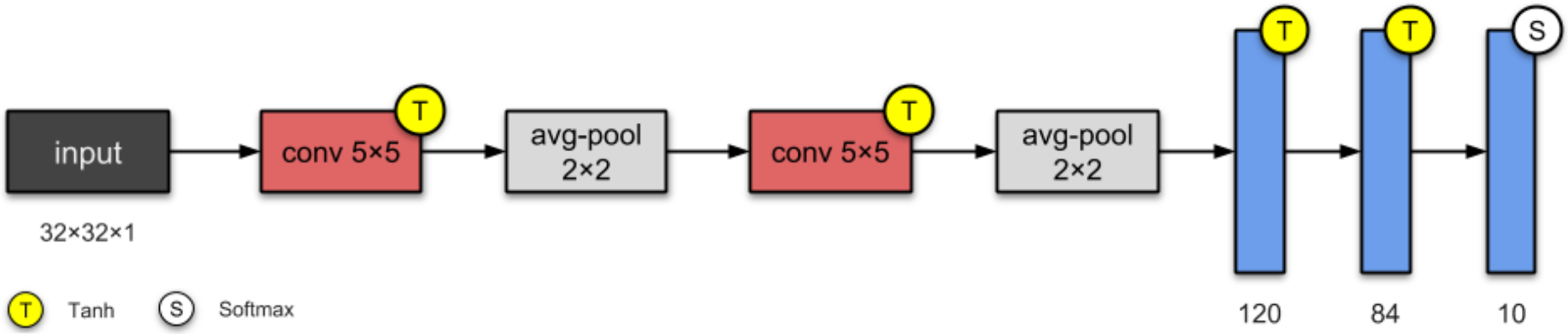
☐ LeNet-5 (1998)

Layer		Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	32x32	-	-	-
1	Convolution	6	28x28	5x5	1	tanh
2	Average Pooling	6	14x14	2x2	2	tanh
3	Convolution	16	10x10	5x5	1	tanh
4	Average Pooling	16	5x5	2x2	2	tanh
5	Convolution	120	1x1	5x5	1	tanh
6	FC	-	84	-	-	tanh
Output	FC	-	10	-	-	softmax



[https://medium.com/@mgazar/lenet-5-in-9-lines-of-code-using-keras-ac99294c8086]

3. Different models for CNN

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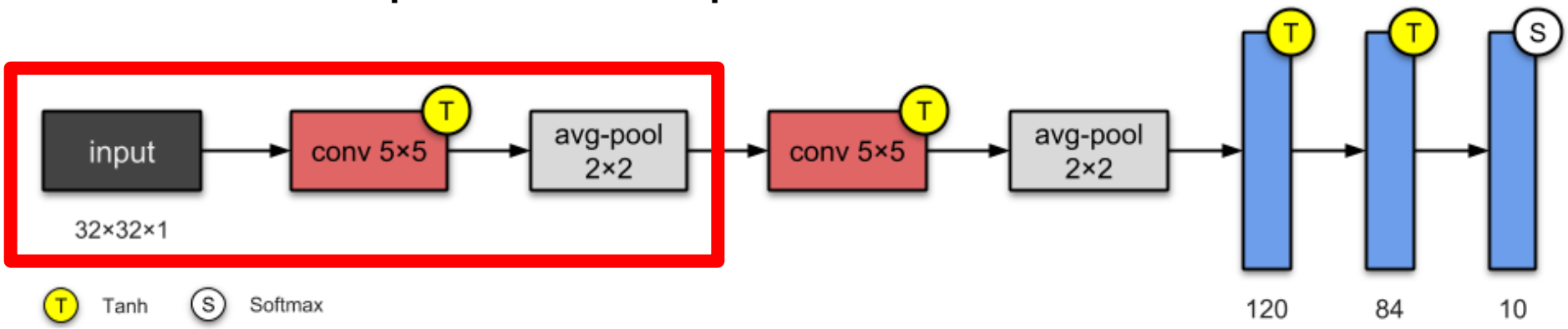
LeNet-5 (1998)

Layer		Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	32x32	-	-	-
1	Convolution	6	28x28	5x5	1	tanh
2	Average Pooling	6	14x14	2x2	2	tanh

- **Conv1**: 6 feature maps (i.e. Edge detection (horizontal, vertical, diagonal), Blob detection), 5×5 kernels, stride 1
  - **Input**: 32×32 grayscale images ➔ **Output**: 28×28×6
- **Avg Pooling1**: 2×2 kernels, stride 2
  - **Input**: 28×28×6 ➔ **Output**: 14×14×6



[https://medium.com/@mgazar/lenet-5-in-9-lines-of-code-using-keras-ac99294c8086]

3. Different models for CNN

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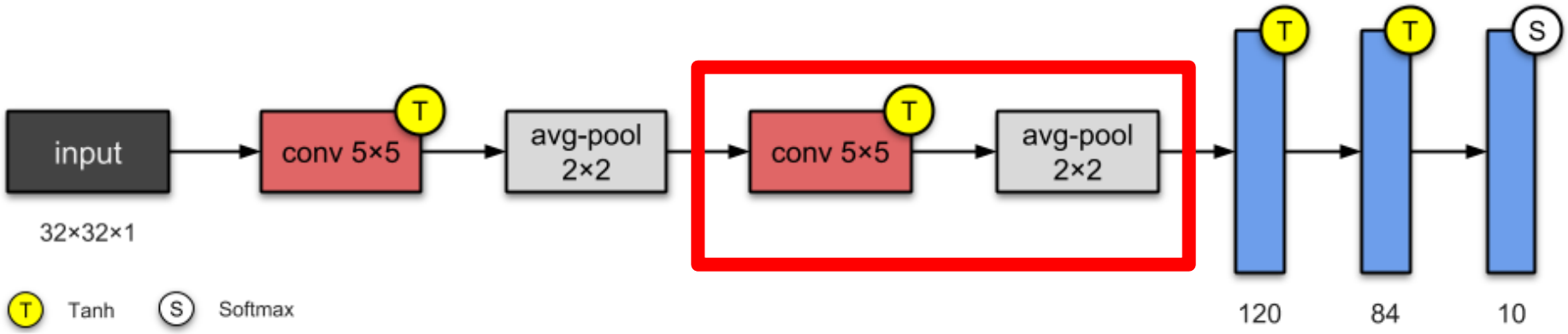
LeNet-5 (1998)

Layer		Feature Map	Size	Kernel Size	Stride	Activation
3	Convolution	16	10x10	5x5	1	tanh
4	Average Pooling	16	5x5	2x2	2	tanh

- **Conv2**: 16 feature maps, 5×5 kernels, stride 1
  - **Input**: 14×14×6 ➔ **Output**: 10×10×16
- **Avg Pooling2**: 2×2 kernels, stride 2
  - **Input**: 10×10×16 ➔ **Output**: 5×5×16

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	X				X	X	X			X	X	X	X		X	X
1	X	X				X	X	X			X	X	X	X		X
2	X	X	X				X	X	X			X		X	X	X
3		X	X	X			X	X	X	X			X		X	X
4			X	X	X			X	X	X	X		X	X		X
5				X	X	X			X	X	X	X		X	X	X



[https://medium.com/@mgazar/lenet-5-in-9-lines-of-code-using-keras-ac99294c8086]

3. Different models for CNN

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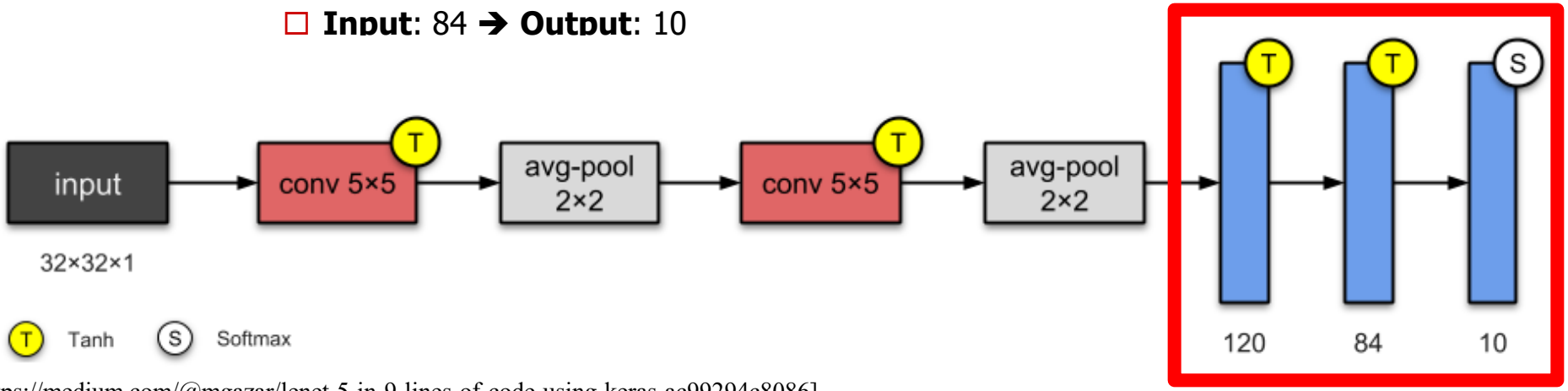
- **LeNet-5 (1998)**

Layer		Feature Map	Size	Kernel Size	Stride	Activation
5	Convolution	120	1x1	5x5	1	tanh
6	FC	-	84	-	-	tanh
Output	FC	-	10	-	-	softmax

- **Conv3**: 120 feature maps (i.e. Intersection patterns), 5×5×16 kernels, stride 1
  - **Input**: 5×5×16 ➔ **Output**: 1×1×120
- **FC1**: Feedforward NN with 84 units
  - **Input**: 1×1×120 ➔ **Output**: 84
- **FC2**: Feedforward NN with 10 units
  - **Input**: 84 ➔ **Output**: 10



[https://medium.com/@mgazar/lenet-5-in-9-lines-of-code-using-keras-ac99294c8086]

4. Dataset for CNN

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- **What are datasets for image classification?**
  - Image classification datasets consist of images and labels.
  - Labels define which class(es) each image belongs to.
  - Each image should have a label since this is a supervised learning task.
  - The labels must be accurate, otherwise the model will pick up the errors in the dataset.

4. Dataset for CNN

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- **Dataset types**
  - **Binary Classification**
    - Two possible classes
    - Model output: Single probability (sigmoid activation).
    - Example: cat vs. dog
  - **Multiclass Classification**
    - More than two mutually exclusive classes
    - Model output: One probability per class (softmax activation).
    - Example: Handwritten digits (1,2,3,4,5,6,7,8,9,0)
  - **Multilabel Classification**
    - Multiple independent labels per sample
    - Model output: Multiple probabilities (sigmoid per class).
    - Example: Image tagging (person, female, Kiwi)

4. Dataset for CNN

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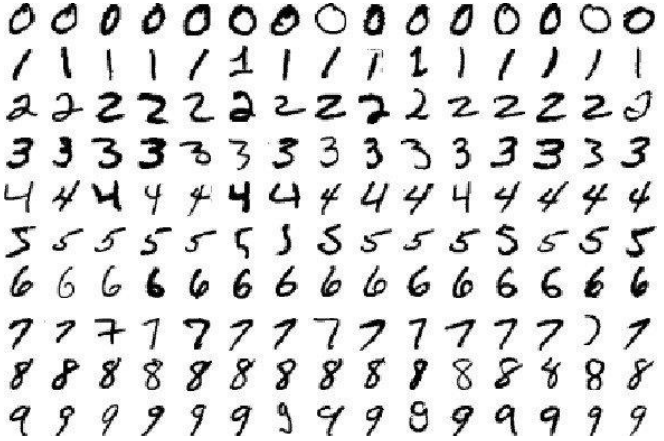
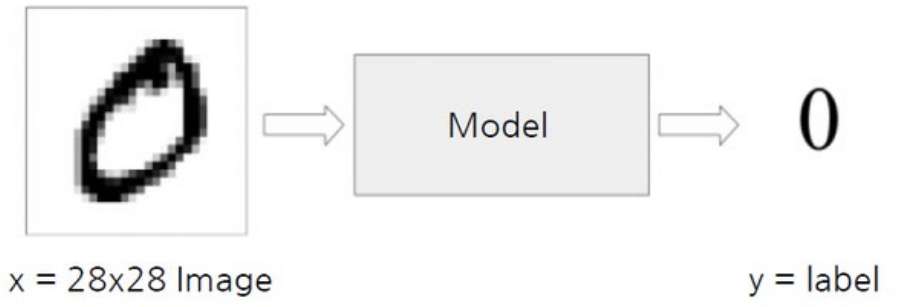
Dataset Example

- **Number recognition: MNIST**
  - Dataset have 0~9 number images
  - 28 x 28 = 784 pixels per image





4. Dataset for CNN

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- **ImageNet**
  - **Multi-class dataset**
  - **Large-Scale Dataset**
    - Contains 14M+ labelled images across 20,000+ categories.
    - Common benchmark for image classification tasks.
  - **Standard Subset for Training & Testing**
    - ImageNet-1K: 1.28M training images, 50K validation images, 1,000 classes.
    - ImageNet-21K: A larger version with 21,841 categories.

4. Dataset for CNN

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- **ImageNet**
  - **ImageNet Challenge (ILSVRC)**
    - ImageNet Large Scale Visual Recognition Challenge (ILSVRC) held annually (2010–2017).
    - Led to breakthroughs in deep learning, e.g., AlexNet (2012), ResNet (2015), ViT (2020).
  - **Dataset Properties**
    - Images are high-resolution and diverse (varied lighting, backgrounds).
    - Classes include animals, objects, scenes, and tools.
  - **Labelling Process**
    - Images in ImageNet are labelled using WordNet synsets (groups of synonyms representing a concept).
    - Each image belongs to a single synset (class).

4. Dataset for CNN

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- **Dataset types**
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5. More on Dataset

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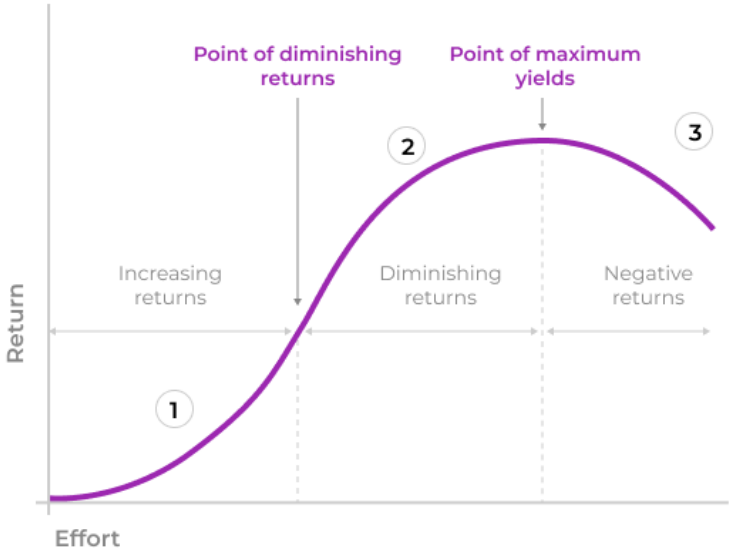
- **Dataset Size vs. Accuracy**
  - **Larger Datasets Improve Accuracy**
    - More training data generally leads to higher classification accuracy.
      - ➔ Better to have larger dataset always??
  - **Computational Cost**
    - Larger datasets require more processing power and training time.
  - **Diminishing Returns**
    - Accuracy gains slow down as dataset size increases.



5. More on Dataset

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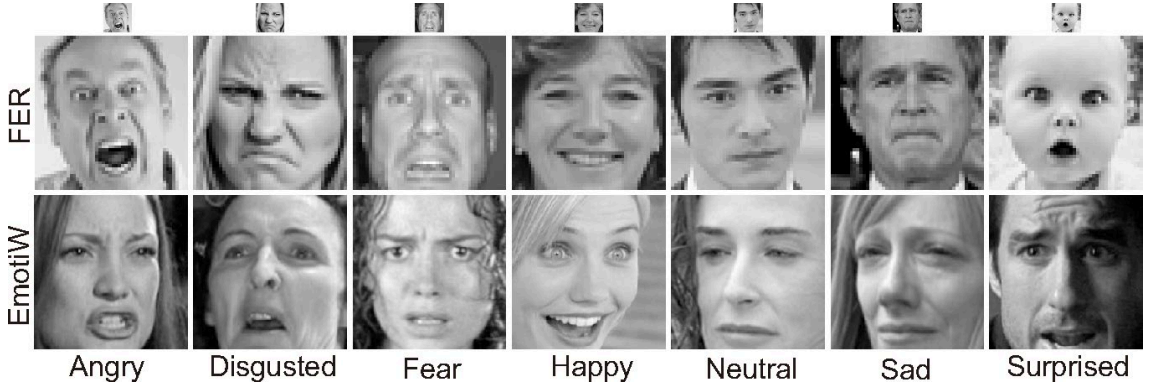
- **Dataset Size vs. Accuracy**
  - **Model Sensitivity**
    - May have different results with different models.
  - **Difference from Datasets**
    - Labelling may be different from Datasets.



  - **Quality Matters**
    - Noisy or poorly labelled data can reduce accuracy, even with a large dataset.

5. More on Dataset

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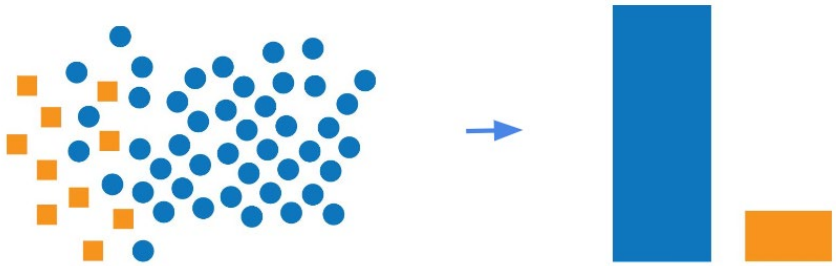
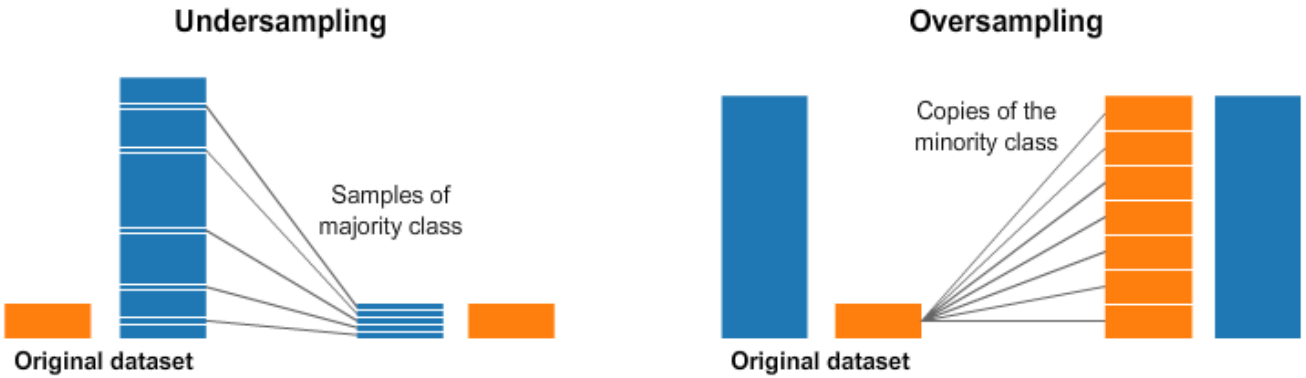
- **Dataset Size vs. Accuracy**
  - **Imbalanced Data**
    - Increasing dataset size may not help minority classes without balancing techniques.



  - **Solution?**



6. Data Augmentation

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Data Augmentation

What is Data Augmentation?

- Technique to artificially expand the training dataset.
- Helps improve model generalization and reduce overfitting.

Why Use Data Augmentation?

- Increases dataset diversity.
- Reduces overfitting, improving generalization.
- Enhances model robustness to real-world variations.
- Useful for small datasets.
- Sometimes can help, but augmentations can hurt accuracy

6. Data Augmentation

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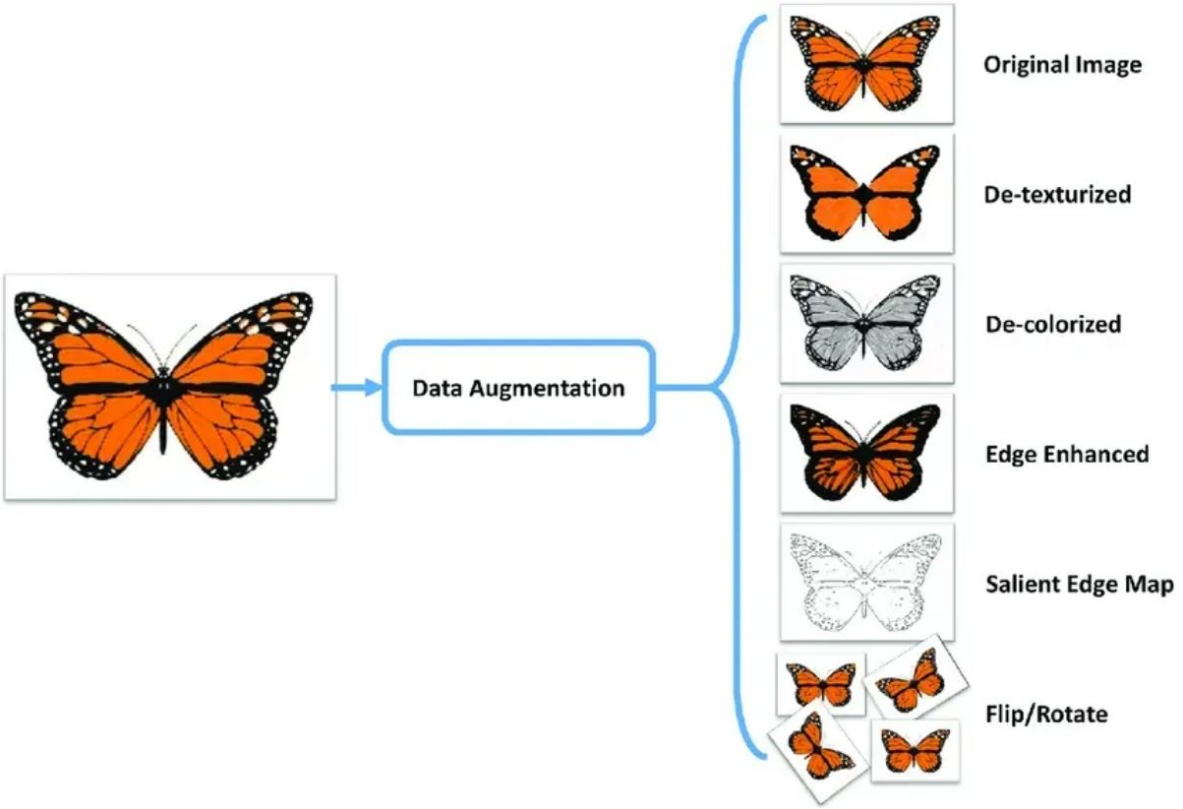
- **Data Augmentation**
  - **Common Augmentation Techniques**
    - Image Data: Rotation, flipping, cropping, scaling, brightness adjustment.



6. Data Augmentation

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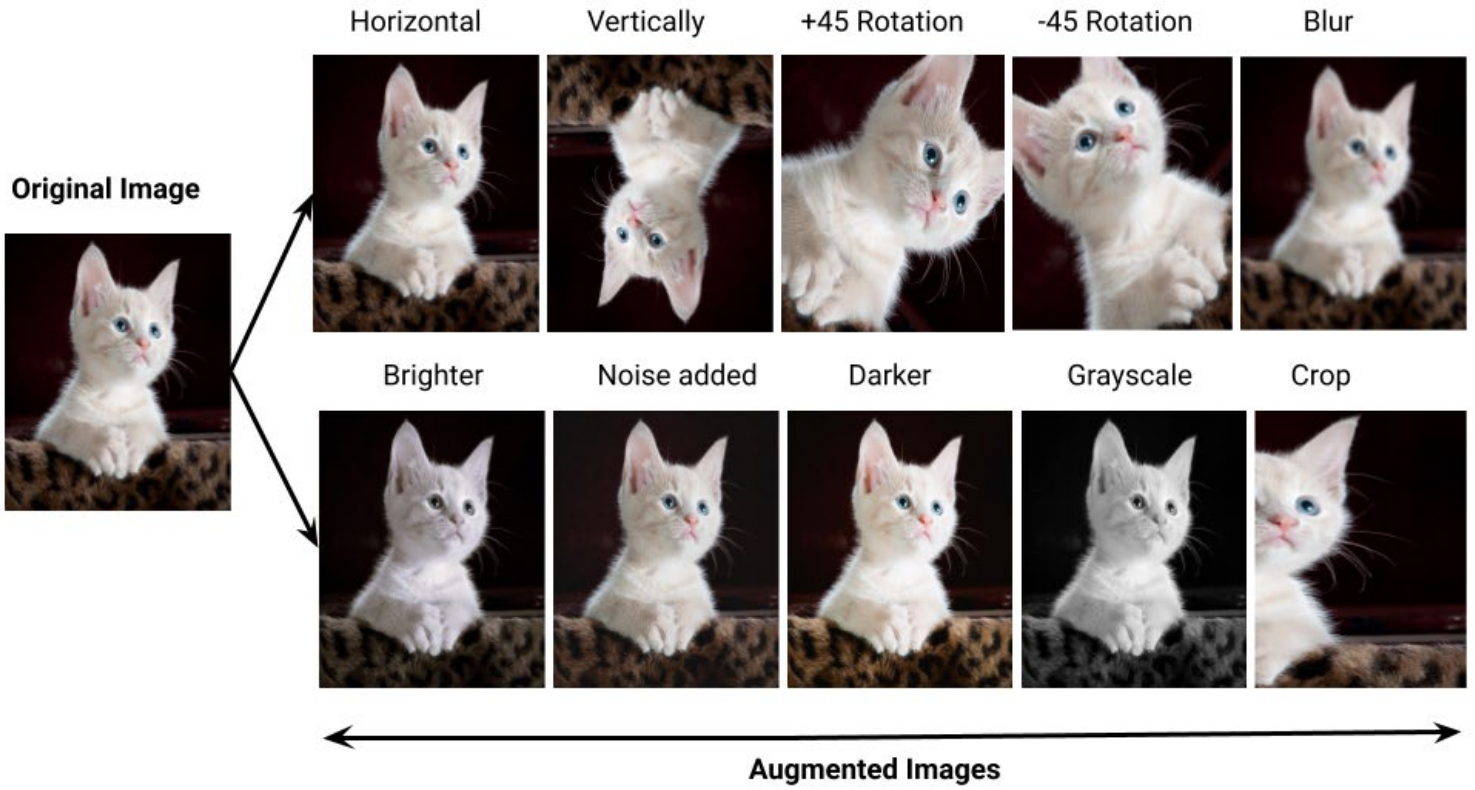
☐ Data Augmentation

☐ Common Augmentation Techniques

- ☐ Image Data: Rotation, flipping, cropping, scaling, brightness adjustment.



6. Data Augmentation

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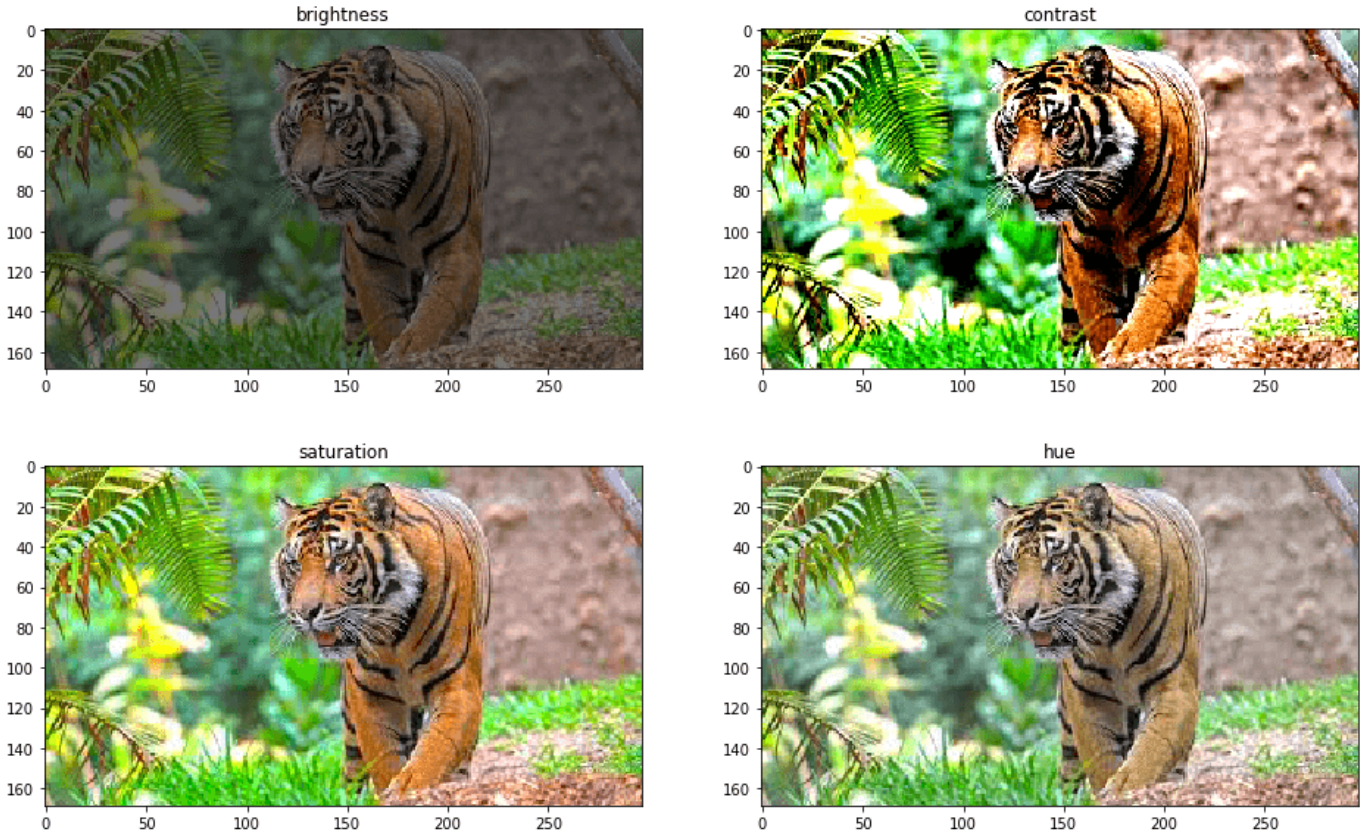
Data Augmentation

Common Augmentation Techniques

- Image Data: Rotation, flipping, cropping, scaling, brightness adjustment.



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☐ Data Augmentation

☐ Pytorch transforms

- `transforms.Compose([])`: Chains multiple image transformations sequentially.
- `RandomResizedCrop(224, scale=(0.8, 1.0))`: Randomly crops and resizes the image to **224×224**, keeping **80-100%** of the original size.
- `RandomHorizontalFlip(p=0.5)`: Flips the image horizontally with a **50% probability**.
- `RandomRotation(degrees=30)`: Rotates the image randomly **within ±30 degrees**.
- `ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1)`: Randomly modifies image **brightness, contrast, saturation, and hue**.
- `ToTensor()`: Converts the image into a **PyTorch tensor** (normalizing pixel values to **[0,1]**).

```python
import torch
import torchvision.transforms as transforms

# Define a sequence of augmentations
transform = transforms.Compose([
    transforms.RandomResizedCrop(224, scale=(0.8, 1.0)),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomRotation(degrees=30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1),
    transforms.ToTensor(),
])
```

Original Image

Augmented Image

6. Data Augmentation

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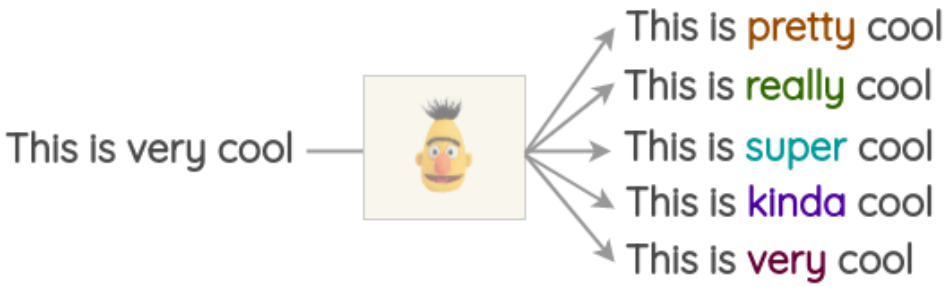
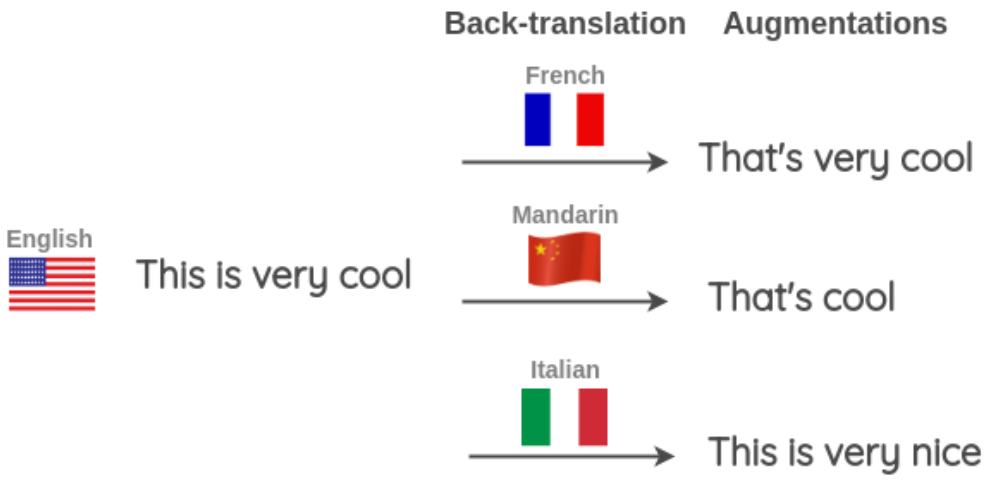
Data Augmentation

Common Augmentation Techniques

- Text Data: Synonym replacement, back translation, word shuffling.





6. Data Augmentation

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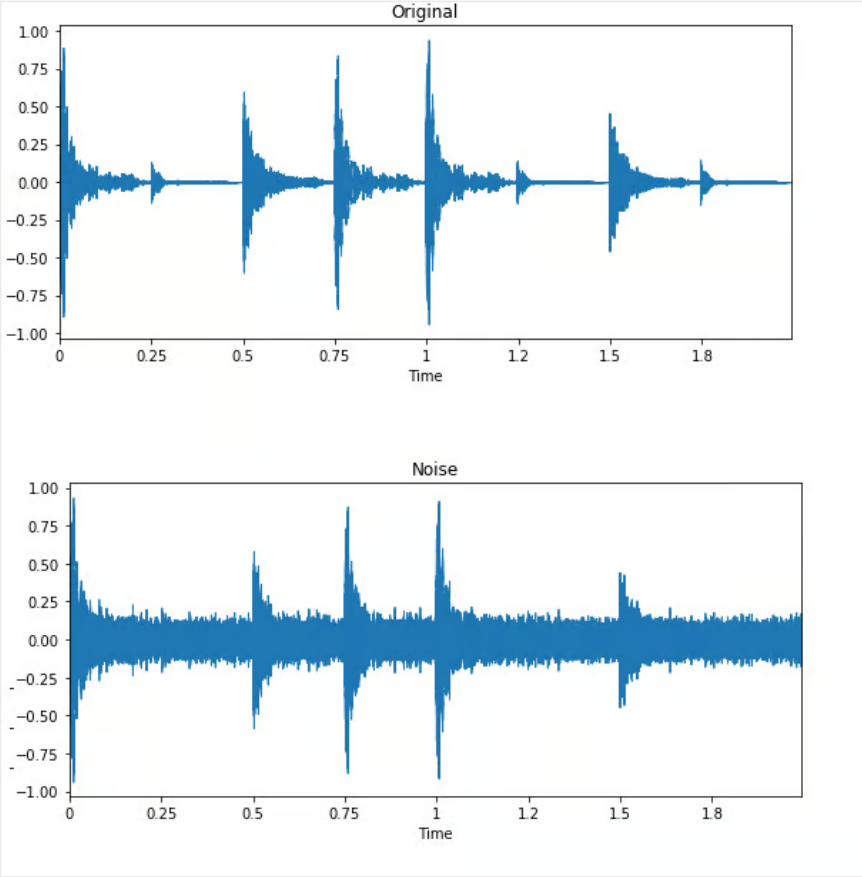
☐ Data Augmentation

☐ Common Augmentation Techniques

- ☐ Audio Data: Time stretching, pitch shifting, noise addition.



6. Data Augmentation

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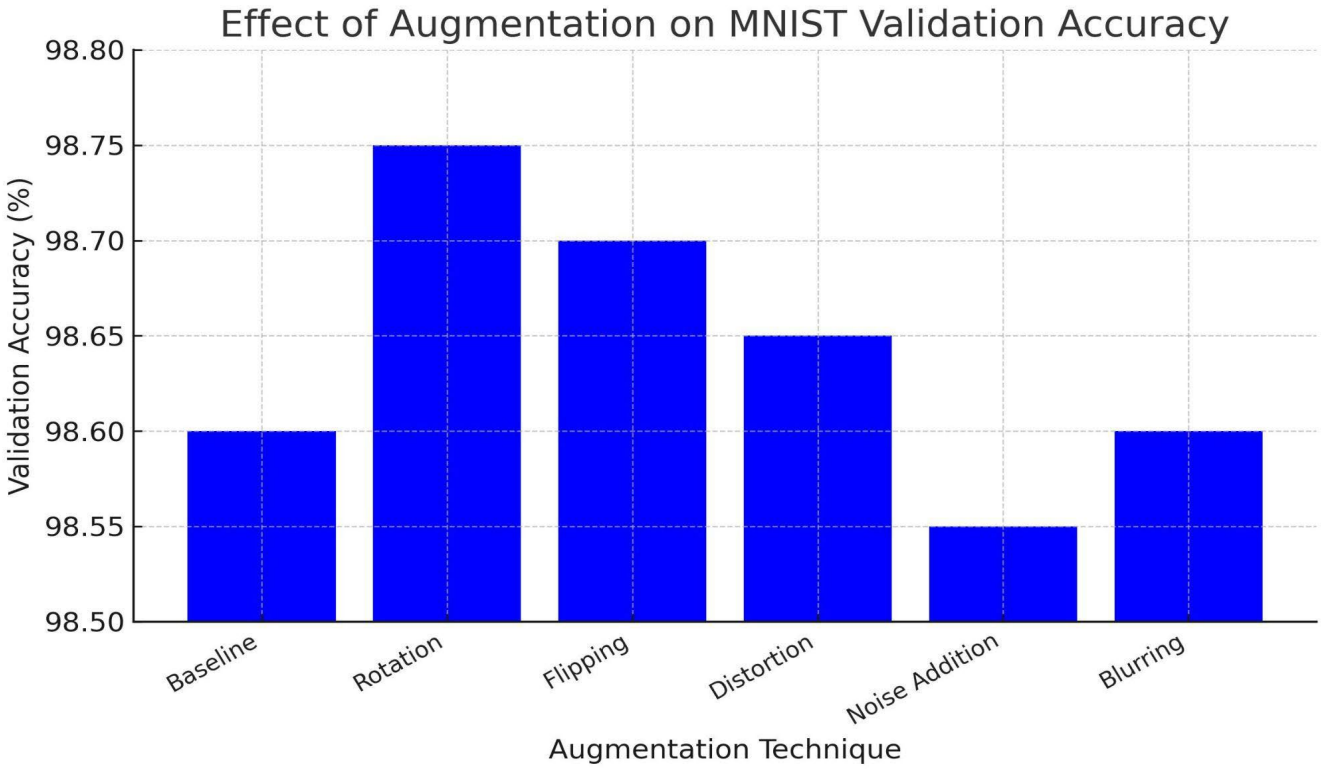
☐ Data Augmentation

☐ Enhancing accuracy



6. Data Augmentation

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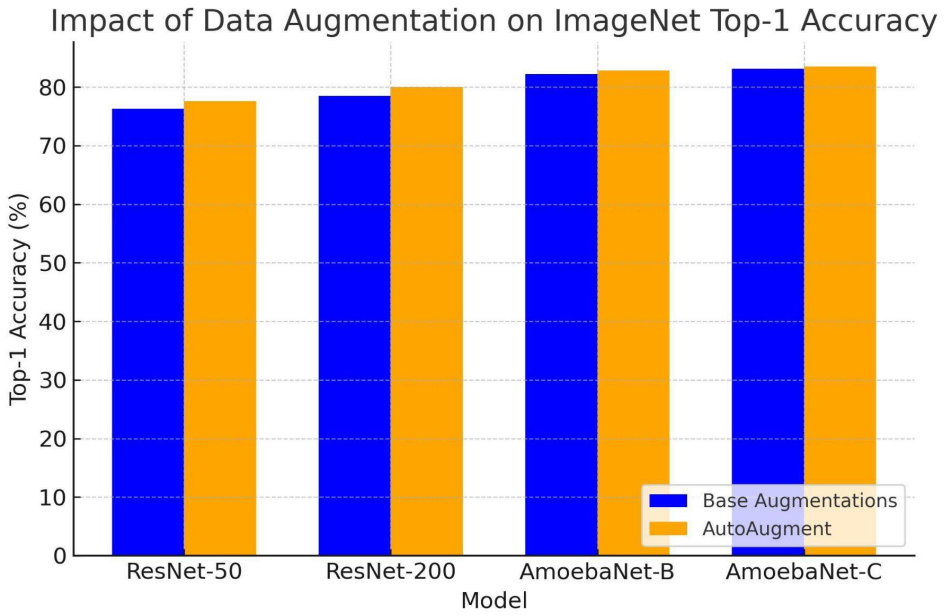
☐ Data Augmentation

- ☐ **AutoAugment Improves Accuracy:**
  - ☐ All models show an accuracy boost with AutoAugment compared to base augmentations.



6. Data Augmentation

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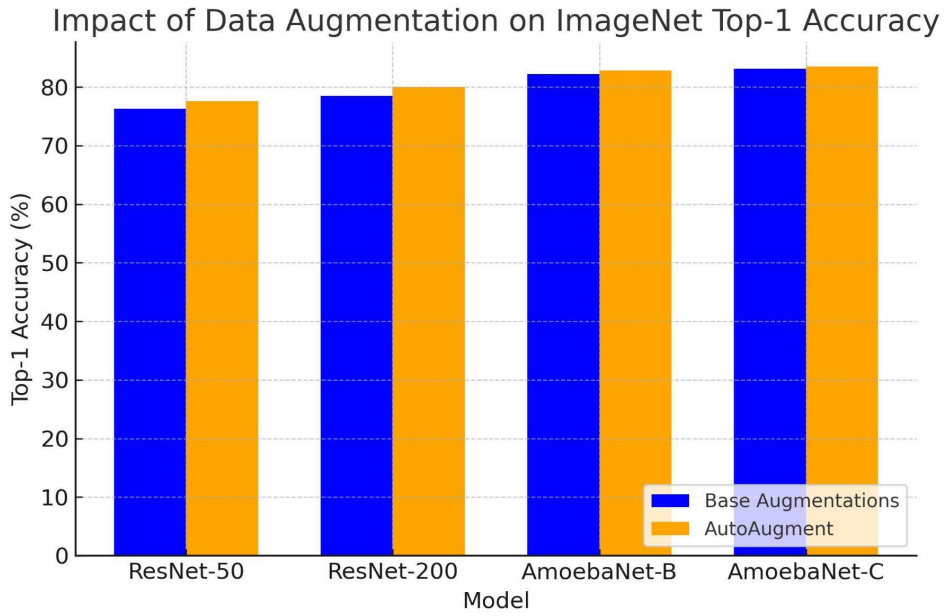
☐ Data Augmentation

- ☐ **ResNet-50 and ResNet-200:**
  - ☐ ResNet-50: Accuracy improves from 76.3% → 77.6% (+1.3%).
  - ☐ ResNet-200: Accuracy improves from 78.5% → 80.0% (+1.5%).



6. Data Augmentation

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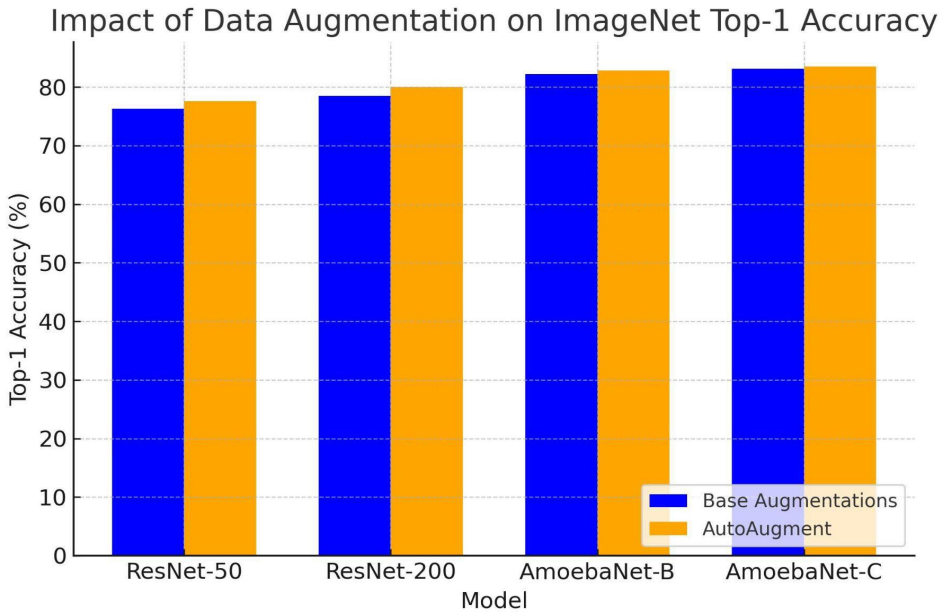
Data Augmentation

- **AmoebaNet Models Show Smaller Gains:**
  - AmoebaNet-B: 82.2% → 82.8% (+0.6%).
  - AmoebaNet-C: 83.1% → 83.5% (+0.4%).



6. Data Augmentation

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Data Augmentation

- **Make sure…**
  - Your augmentations should not change the data so much that your classes become unrecognizable.
  - You visualize your augmentations and check it.

- **Which augmentations should you use?**
  - If your model is performing well already - None
  - If your model is struggling - try add some simple augmentations
  - If your model is still performing poorly - try advance augmentations

➔ There is no "right" augmentation! It depends on your dataset and model!!!

7. Dataset Splits

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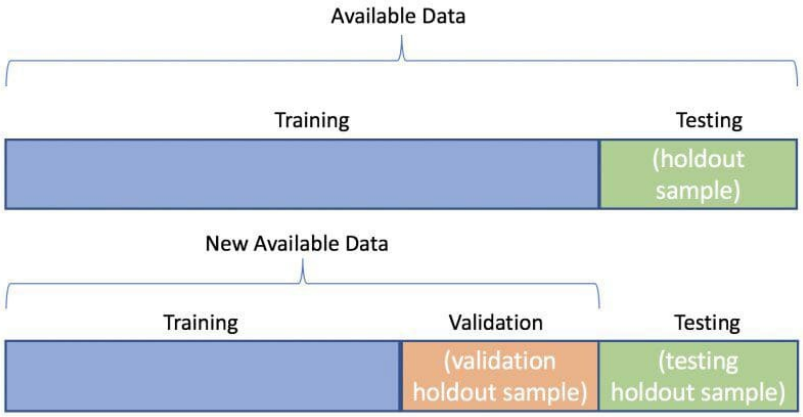
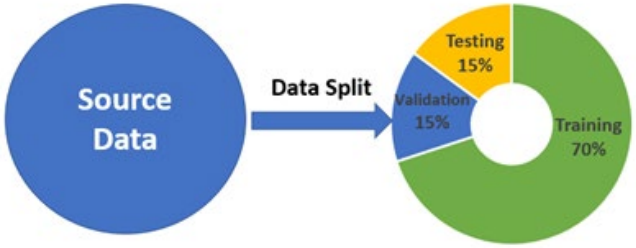
- **Dataset Splits**
  - **Training Set**
    - Largest split (~70-80% of data).
    - Used for model learning and parameter updates.
  - **Validation Set**
    - Smaller split (~10-20% of data).
    - Used for hyperparameter tuning and early stopping.
  - **Test Set**
    - Final split (~10-20% of data).
    - Evaluates generalization on unseen data.





7. Dataset Splits

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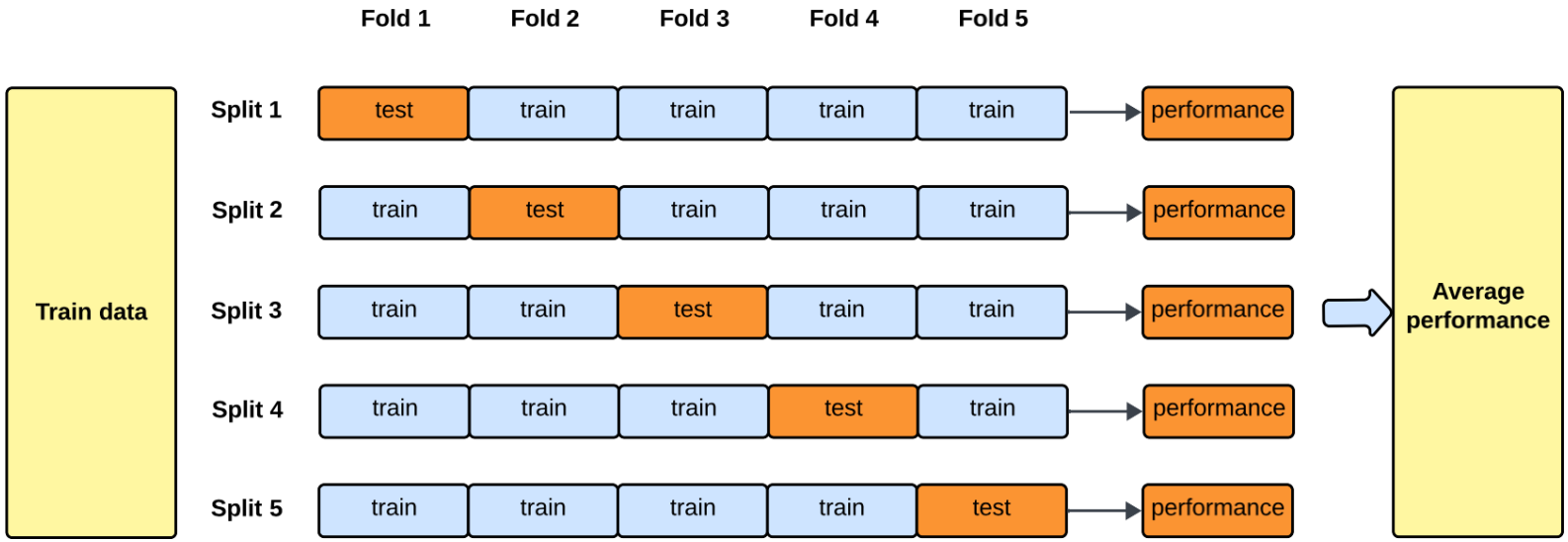
Dataset Splits

Training and Test Sets

- Usually 80% - 20%.



7. Dataset Splits

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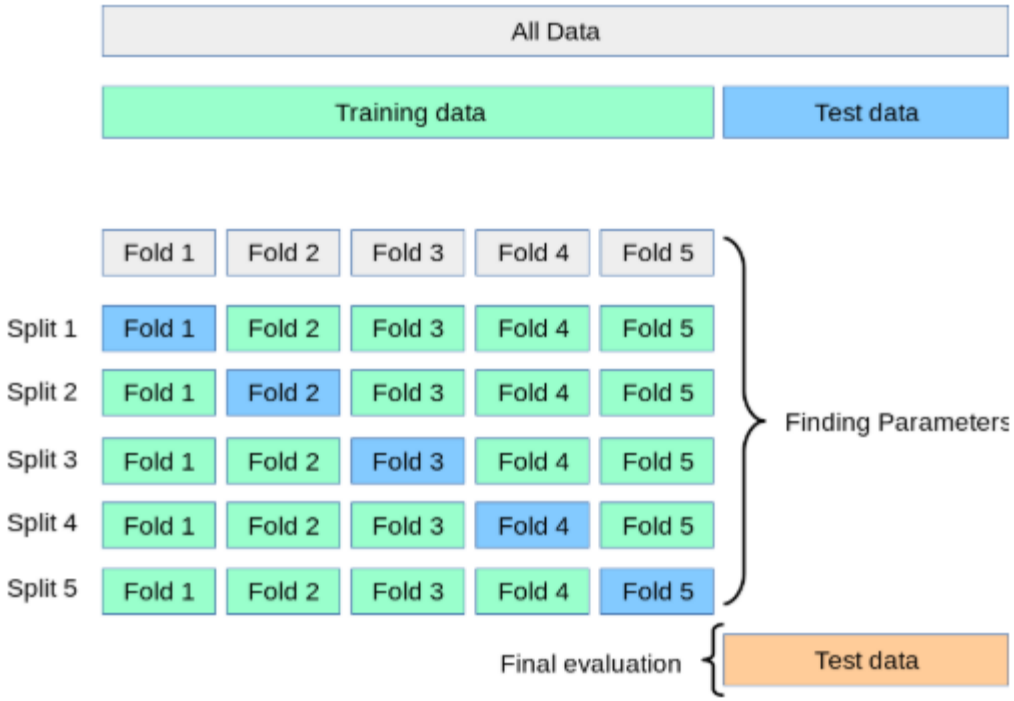
Dataset Splits

Training, Validation and Test Sets

- Usually 60% - 20% - 20%.



8. Summary

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- **More training or data/augmentation needed?**
  - **If your model is undertrained:**
    - If you have only trained for a few epochs and your validation accuracy is still improving, **continue training for longer** before making other adjustments.
  - **If validation accuracy is stuck after many epochs:**
    - If you have trained for a large number of epochs but validation accuracy is not improving, **try adding more data augmentation** to enhance generalization.
  - **If multiple augmentations do not help:**
    - If you have already applied multiple augmentations and validation accuracy is still not improving, you **may need more data** to improve performance.
  - **Training strategy recommendations:**
    - Aim to **train your model first** until it converges (i.e., stops improving) before introducing additional augmentations.
    - Try **adding augmentations** before increasing the dataset size to maximize data efficiency.

8. Summary

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- **Dataset Quality Matters** – Labels must be accurate to avoid training on incorrect information.
- **Dataset Size vs. Accuracy** – More data generally improves accuracy, but with diminishing returns.
- **Proper Dataset Splits** – Train (~70-80%), validation (~10-20%), and test (~10-20%) ensure reliable evaluation.
- **Data Augmentation Helps** – Increases diversity, reduces overfitting, and improves generalization.
- **Common Augmentations** – Rotation, flipping, cropping, scaling, brightness adjustment, noise.
- **Advanced Techniques Boost Performance** – MixUp, CutMix, GridMask, AutoAugment, and AugMix.
- **Augmentation vs. Accuracy Trade-off** – Helps but may not always improve performance; requires tuning.
- **No Universal Best Augmentation** – Effectiveness depends on dataset, model, and task requirements.
- **Experimentation is Key** – Test different augmentations to find the best combination for your model.